

The Open Source Way to EU Digital Sovereignty & Competitiveness

A Detailed Roadmap for Action

*Based on the July 2025 Report by the European
Alliance for Industrial Data, Edge and Cloud*

Part 1: The Strategic Imperative

Why Europe Must Act Now

This section outlines the foundational concepts and the urgent drivers behind the strategic push for digital sovereignty, powered by Open Source.

What is Digital Sovereignty?

"The capacity for autonomous assessment, decision-making, and action in the digital space."

To simplify, it is a combination of two key elements:

Digital Sovereignty = Data Sovereignty + Technological Autonomy

This means not only controlling our data but also the underlying hardware and software that manages it. This autonomy is essential for safeguarding Europe's operational independence.

Four Key Motivations Driving the Roadmap

The report is driven by four urgent motivators for Europe:

1. **Digital & Technological Sovereignty**

Asserting control over our digital infrastructure and reducing reliance on non-EU technologies.

2. **Data Security & Compliance**

Safeguarding data and privacy with transparent, auditable solutions that comply with European laws like GDPR and the NIS Directive.

3. **Innovation & Economic Resilience**

Fostering a competitive and innovative market by lowering entry barriers for SMEs and promoting collaborative development.

4. **Environmental Sustainability**

Reducing the environmental footprint of digital infrastructure, in line with the European Green Deal.

Open Source: The Cornerstone of Sovereignty

More than a matter of cost, Open Source is a fundamental enabler of technological sovereignty.

- **Transparency & Security**

The ability to audit the code allows for collective verification, finding and fixing vulnerabilities, and ensuring there are no hidden backdoors.

- **Autonomy & Reversibility**

It offers an escape from vendor lock-in. Full control over the code guarantees the freedom to adapt, modify, and innovate independently.

- **Collaboration & Innovation**

Open Source allows for pooling resources to build cutting-edge technologies that no single entity could develop alone, fostering a vibrant ecosystem.

Part 2: The Landscape & The Gaps

Europe's Current Position and Critical Challenges

This section assesses the current European Open Source ecosystem and identifies the five major obstacles hindering the achievement of digital sovereignty.

A Shared, Yet Challenged, European Ambition

Digital sovereignty is a strategic goal for Europe, supported by key stakeholders, but significant challenges remain.

Key Initiatives:

- **European Alliance for Industrial Data, Edge and Cloud:** A coalition of industry, Member States, and experts.
- **Next Generation Internet (NGI):** A European Commission program to build a more human-centric internet.
- **IPCEI-CIS:** An "Important Project of Common European Interest" for Cloud Infrastructure and Services.

The Reality:

Despite these efforts, Europe's digital economy is heavily reliant on non-European entities. The IT market is still largely dominated by non-EU proprietary technologies, especially from US-based hyperscalers.

The 5 Critical Gaps Holding Europe Back

The report identifies five interconnected gaps that prevent the full realization of a sovereign digital infrastructure.

1. **Lack of open standards and interoperability**
2. **Under-investment in critical software components**
3. **Barriers to adoption (market, visibility)**
4. **Shortage of talent and training**
5. **Project governance often outside the EU**

Addressing these gaps requires the **70 concrete actions** proposed in the roadmap.

Part 3: The Roadmap for Action

The 5 Strategic Pillars and Key Proposals

This section details the core of the roadmap: a five-pillar strategy with actionable recommendations to build a sovereign, resilient, and competitive digital Europe.

The 5 Strategic Pillars of the Roadmap

The 70 proposals are organized across five pillars, forming a comprehensive strategy for action:

- **Pillar 1: Technological Development**

Build a solid foundation on open, interoperable standards.

- **Pillar 2: Skills Development**

Train the talent needed to build and maintain the ecosystem.

- **Pillar 3: Public Procurement**

Use public purchasing power as a strategic lever for change.

- **Pillar 4: Growth and Investment**

Create a sustainable funding environment for European Open Source.

- **Pillar 5: Governance**

Ensure the long-term sustainability and European control of critical projects.

Pillar 1: Technological Development

Objective: To build a resilient technical foundation based on truly open standards and well-supported European projects.

Key Actions:

1. Establish & Enforce TRULY Open Standards

- Develop European-governed specifications and reference architectures to fight "open washing" and ensure genuine interoperability.

2. Fund Critical Open Source Projects

- Create a "**European Open Source Sovereignty Fund**" (EOSSF) to provide stable, long-term funding for essential infrastructure components.

3. Promote EU Reference Implementations

- Develop sector-specific "playbooks" and large-scale demonstration projects (e.g., in healthcare, public administration) to showcase European excellence.

Pillar 2: Skills Development

Objective: To bridge the talent gap by fostering a skilled workforce proficient in European Open Source technologies.

Key Actions:

1. Launch Targeted Training & Certifications

- Create industry-recognized certifications for European Open Source technologies, focusing on practical, hands-on skills.

2. Invest in Reskilling & Upskilling

- Fund programs to help IT professionals transition to roles in the European Open Source ecosystem.

3. Integrate Open Source into Education

- Incorporate Open Source principles into STEM curricula and establish "Centres of Excellence" at universities.

Pillar 3: Public Procurement

Objective: To leverage public spending as a powerful catalyst for the adoption of European Open Source solutions.

Key Actions:

1. Adopt "Public Money, Public Code, European Preference"

- Mandate that taxpayer-funded software originates from and benefits the European Open Source ecosystem.

2. Develop Clear Procurement Guidelines

- Provide resources for public bodies to evaluate and select sovereign Open Source solutions, focusing on total cost of ownership (TCO) and avoiding vendor lock-in.

3. Use Pre-Commercial Procurement (PCP)

- Co-develop innovative solutions with European SMEs to address specific public 13

Pillar 4: Growth and Investment

Objective: To build a robust and sustainable funding ecosystem for the growth of European Open Source.

Key Actions:

1. Create a Centralized Investment Platform (EOSIP)

- Develop a one-stop-shop platform consolidating information on grants, loans, and venture capital for European Open Source projects.

2. Establish a "European Open Source Brand"

- Launch a branding initiative to promote the quality, security, and reliability of sovereign European solutions on the global stage.

3. Foster Public-Private Partnerships (PPPs)

- Incentivize private sector investment in critical Open Source projects through co-investment programs and R&D consortia.

Pillar 5: Governance

Objective: To ensure the long-term security, sustainability, and European control of critical Open Source projects.

Key Actions:

1. Conduct Proactive Risk & Vulnerability Analysis

- Partner with EU cybersecurity agencies to assess critical projects and reduce dependency on non-EU components.

2. Align with Existing EU Regulations

- Provide guidance to help European projects navigate and comply with frameworks like the Cyber Resilience Act (CRA) and GDPR without undue burden.

3. Strengthen European Leadership

- Establish a **European Open Source Advisory Board** to oversee funding,

Part 4: The Added Value & Vision

The Tangible Benefits and a Framework for the Future

This section highlights the concrete benefits of implementing the roadmap across key sectors and introduces a framework for evaluating true sovereignty.

Transformative Impact on Key Sectors

Adopting European Open Source solutions will unlock significant benefits:

- **Public Administration**

Enhances security, transparency, and accountability while reducing dependency on non-EU providers and lowering costs.

- **Manufacturing (Industry 4.0)**

Enables predictive maintenance, improves interoperability on the factory floor, and fosters a culture of innovation.

- **Healthcare**

Ensures patient data security and privacy, enables interoperability between systems, and accelerates medical research through secure collaboration.

- **Energy**

Supports a greener future by optimizing energy grids, integrating renewables, and reducing the energy consumption of data centers.

"LOTEC" Criteria for Effective Sovereignty

To assess and ensure true digital sovereignty, the report proposes a multi-dimensional evaluation framework (work in progress):

- **L - Legal:** Immunity from non-EU laws (e.g., CLOUD Act). The supplier and its beneficial owners must be under EU jurisdiction.
- **O - Operational:** Control of operations (admin, support) by European personnel located in Europe.
- **T - Technological:** Transparency, interoperability, and reversibility guaranteed by **Open Source** and open standards.
- **E - Economic:** Value creation (R&D, jobs) primarily in Europe with fair business models.
- **C - Cultural & Ethical:** Alignment with European values and diversity.

Conclusion: A Call to Action

European digital sovereignty is not a distant goal—it is an urgent necessity that can be achieved through collective, strategic action. **Open Source is Europe's greatest asset in this endeavor.**

The time for decisive action is now.

Get the full report:

<https://ec.europa.eu/newsroom/dae/redirection/document/117980>